range to have hands-on experience and learn things the practical way. Built entirely with mud from under the feet and powered by the sun over the head, this is one of the first schools that has had a zero energy off-grid campus.

With abundant sunshine available almost up to 300 days a year, solar-heated mud houses have been built in Ladakh. Despite their low cost of construction, these houses are efficient enough to provide +18°C environment in a -20°C winter, with nothing but the sun to help.

Himalayan Institute of Alternatives, Ladakh is also working on some local solutions. "One of the interesting experiments is where we've built four passive solar-heated (PSH) mud buildings and obtained fascinating results," according to them.

The graphs in Fig. 2 compare the difference in temperature for a PSH house, a fossil fuel heated building, and those buildings that are not heated. They are represented by the orange, maroon, and bluish-grey lines, respectively. With the month of January being the coldest, the average temperature inside the solar-powered houses ranges from +18 to +25°C, even though the temperature outside is -15°C.

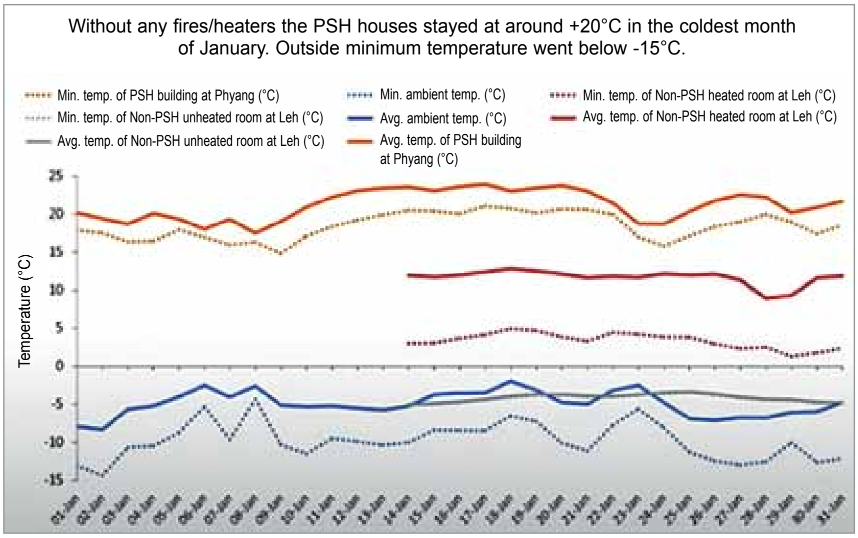

Fig. 2: How PSH houses are more effective than the others

Fig. 3: Working of the mountain-top internet using line-of-sight propagation

Construction

The PSH buildings are built using straw and clay. The latter is quite abundant in Ladakh, while the former comes from the state of Punjab, where it is seen as a source of pollution when burnt. Sonam says, "We rescue it (straw) and save it from polluting Punjab and Delhi. When the clay and straw are mixed, they become lightweight insulated blocks that are used in Ladakh to build the solar-powered houses, thus cutting pollution in both Punjab and Ladakh."

The buildings are generally oriented towards the south where the winter sun roams most of the time. Behind the glazing of the southern wall, a dark wall is erected for absorbing the light waves (and the heat), thus capturing the solar energy, and storing it in a sort of no-cost battery that is made of plastic waste bottles. The water in the plastic bottles traps the heat during daytime and releases this heat at night.

To retain the precious heat indoors, a good amount of insulation is required in the three other side walls as well as the roof and floor. This insulating material is made from the wood and wool waste obtained from the Pashmina industry. Their latest experiment is using solar-powered mud principles along with RCC frame structures to obtain the conventional benefits along with solar passive heating.

The amount of oil burned on the Indian borders of China and Pakistan is immense. It is used for heating the army shelters at around 3500 to 4575 metres (11,500 to 15,000 feet) altitude. However, it produces 330,000 tonnes of carbon-dioxide. These are now going solar, with the shelters being redesigned to maintain the room temperature between +12 and +15°C, when the outside world is at -22 to -30°C.

Ice Stupa artificial glacier

Sonam says, "With global warming melting away glaciers, people laughed when we spoke of artificial glaciers." The principle is to freeze the stream of winter water that goes unused as there is no farming in that area. A pipe is placed upstream and brought down to where the fields are. Since water tries to flow downwards through the pipe, the water gushes out at lower end of the pipe without any power used. Because of the -20°C or so temperature outside, the water starts freezing in the form of a cone below the pipe, thus forming an Ice Stupa.